

Directed Acyclic Graphs

BUSS975: Causal Inference in Financial Research

Professor Ji-Woong Chung

Korea University

Outline

1. Why graphs?
2. DAG notation
3. Three elemental structures
4. Backdoor paths and adjustment
5. Collider bias
6. Bad controls
7. Neutral controls and precision
8. What DAGs cannot do

1. Why Graphs?

Where we left off

- Last lecture: a causal effect is a property of the data **plus the assignment mechanism**.
- The potential outcomes framework gave us:
 - ▶ precise **estimands** — ATE , ATT , ATU ;
 - ▶ a decomposition of what goes wrong — selection bias and heterogeneity bias;
 - ▶ a gold standard — randomization, which makes $(Y^1, Y^0) \perp\!\!\!\perp D$.
- What it did **not** give us: a systematic way to answer the single most common question in empirical work.

“What should I control for?”

The question every referee asks

- A typical referee report: *“The authors should control for firm size, leverage, industry, and profitability.”*
- Is that good advice? The honest answer in most papers is: **nobody knows**.
- Common (bad) heuristics:
 - ▶ “Control for everything you have.” — Wrong, and this lecture shows why.
 - ▶ “Control for anything correlated with both D and Y .” — Closer, but still wrong.
 - ▶ “More controls make the estimate more credible.” — Emphatically wrong.
- Adding a control can **remove** bias, **add** bias, or **flip the sign** of your estimate.
- To tell which, you need a language for your assumptions about the data-generating process.

What a DAG is for

- A **directed acyclic graph** is a picture of your causal assumptions.
- Once drawn, it delivers — mechanically, by rules you can check —
 - ▶ which variables you **must** control for,
 - ▶ which variables you **must not** control for,
 - ▶ whether your causal effect is identified **at all** from observational data.
- The DAG does not tell you whether your assumptions are *true*. Nothing does.
- What it does is make them **explicit, communicable, and refutable** — so a referee can disagree with your *assumptions* rather than your *taste in controls*.

This lecture . . .

By the end you will be able to explain, precisely, why the following is true:

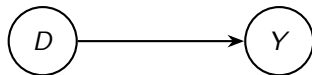
You can run a perfectly randomized experiment, add one perfectly measured control variable, and get an estimate with the wrong sign.

- If controlling for things can do that, “control for everything” is not a strategy — it is a coin flip.

2. DAG Notation

Nodes and arrows

- A DAG has **nodes** (random variables) and **directed edges** (arrows).
- An arrow $X \rightarrow Y$ means: X exerts a **direct causal effect** on Y — direct meaning *not* routed through any other variable in the graph.



- Read: relationship lending (D) affects firm investment (Y).
- The graph is **non-parametric**: it says nothing about magnitude, sign, linearity, or functional form. Only *whether* an effect exists.

The missing arrow is the assumption

- Drawing an arrow is a weak claim: “ X may affect Y .”
- **Omitting** an arrow is a strong claim: “ X has *no* effect on Y , for any unit, of any size.”
- This is the opposite of how we usually read a regression table, and it is the most important thing to remember about DAGs.
- So the interesting content of a DAG is in the arrows you **did not draw**.
- Corollary: a DAG with arrows everywhere is honest but useless — nothing is identified. Progress requires exclusions, and exclusions require **institutional knowledge**.

Acyclicity

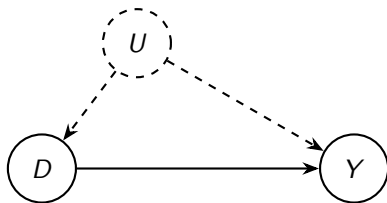
- “Acyclic” = no directed path leads from a variable back to itself.
- Forbidden:



- This is a real restriction for finance, where equilibrium feedback is everywhere: leverage and performance, price and demand, coverage and returns.
- The usual fix is **time**: $\text{Leverage}_t \rightarrow \text{Perf}_{t+1} \rightarrow \text{Leverage}_{t+2}$ is acyclic.
- Genuine simultaneity is beyond what a DAG can express. We return to this in §8, What DAGs Cannot Do.

Observed and unobserved variables

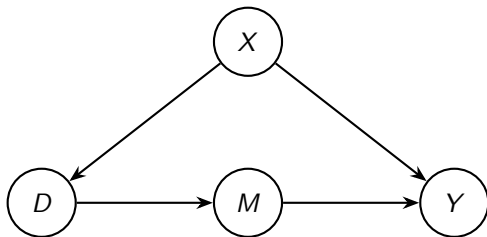
- Solid circle = **observed**; dashed circle = **unobserved**.



- U might be managerial talent, firm culture, private information, or local demand.
- The distinction matters enormously: a variable you cannot measure cannot be conditioned on, so it cannot be used to block anything.
- Much of this course is about what to do when a U like this exists.

Paths

- A **path** is any sequence of edges connecting two nodes, *ignoring the direction of the arrows*.
- A **directed (causal) path** follows the arrows from start to finish.



- $D \rightarrow M \rightarrow Y$ is a directed path (this is the causal effect).
- $D \leftarrow X \rightarrow Y$ is a path but **not** a directed path — yet it still generates correlation between D and Y .
- **Association flows along paths regardless of arrow direction. Causation flows only along directed paths.**

Where DAGs come from

- A DAG is **not** derived from the data. The data cannot tell you which way the arrows point.
- It comes from: economic theory, institutional detail, prior empirical work, and knowledge of how *your specific dataset* was assembled.
- Two researchers with different institutional knowledge will draw different DAGs and reach different conclusions — and that is the point. The disagreement is now **visible and arguable**.
- Practical advice: draw the DAG *before* you run the regression. If you cannot draw it, you do not yet understand your identification strategy.

Conditioning is not intervening

- **Conditioning on X** filters: “among firms that *happen* to have $X = x$.” We keep a subpopulation, which may be special in ways we cannot see.
- **Intervening on X** sets: “if we *assigned* $X = x$ to every firm.” The population stays fixed; only X and its consequences move.
- Prediction needs the first. Causal inference needs the second.
- In graph terms, intervening **deletes every arrow into X** : X is no longer caused by its parents; it is caused by us.
 - ▶ That is what randomization does physically. No arrows in, nothing to block — no controls needed for identification.
- The rest of the machinery answers one question: **when does conditioning on the right Z reproduce intervening?**

Controlling for is not setting

- **Conditioning on X** filters: “among firms that *happen* to have $X = x$.” We keep a subpopulation, which may be special in ways we cannot see.
- **Intervening on X** sets: “if we *assigned* $X = x$ to every firm.” The population stays fixed; only X and its consequences move.
- Prediction needs the first. Causal inference needs the second.
- In graph terms, intervening **deletes every arrow into X** : X is no longer caused by its parents; it is caused by us.
 - ▶ That is what randomization does physically. No arrows in, nothing to block — no controls needed for identification.
- The rest of the machinery answers one question: **when does conditioning on the right Z reproduce intervening?**

Ratings predict default; ratings do not prevent it

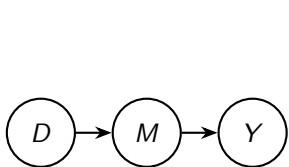
- Let $Y = 1$ if a firm defaults within five years, and let X be its credit rating.
- **Conditioning**: among AAA issuers, default is rare. Ratings are superb *predictors* — selling this number is a profitable business.
- **Intervening**: suppose a regulator relabelled every junk issuer AAA. Default rates would barely move. The label is not the cash flows.
- Conditioning **selects** firms that were already creditworthy; Intervening **changes the label** and leaves creditworthiness untouched.¹
- Ask of any empirical result: *is this a filter, or an intervention?*

¹The causal effect is not exactly zero. Kisgen and Strahan 2010: when the SEC certified DBRS in 2003, yields moved 39bp per notch — the label alone, with fundamentals unchanged.

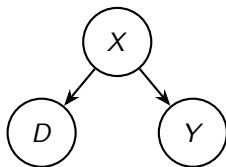
3. Three Elemental Structures

Every DAG is built from three pieces

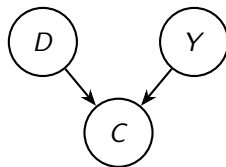
Any path, however long, is a chain of three-node segments of exactly three types.



Chain (mediator)



Fork (confounder)



Collider

Chain: $D \rightarrow M \rightarrow Y$

- M is a **mediator**: the causal effect of D travels *through* it.
- The path is **open** by default — D and Y are correlated, and the correlation is causal.
- Conditioning on M **closes** the path — and thereby destroys the very effect you are trying to measure.
- Example: a credit shock raises investment, which raises employment.
 - ▶ Credit $\rightarrow$ Investment $\rightarrow$ Employment.
 - ▶ Controlling for investment when estimating the effect of credit on employment removes most of the answer.
- **Do not control for mediators** when you want the total effect.

Fork: $D \leftarrow X \rightarrow Y$

- X is a **confounder**: a common cause of both treatment and outcome.
- The path is **open** by default — D and Y are correlated, but the correlation is **not** causal.
- Conditioning on X **closes** the path and removes the bias.
- This is exactly omitted variable bias from last lecture, drawn as a picture.
- Example: firm quality drives both whether a firm has a lending relationship and how much it invests.
- **Do control for confounders** — if you can measure them.

Collider: $D \rightarrow C \leftarrow Y$

- C is a **collider**: a common *effect* of treatment and outcome.
- The path is **closed** by default — D and Y are *not* correlated through it.
- Conditioning on C **opens** the path and *creates* a correlation that did not exist.
- This is the one with no counterpart in your regression training, and it is where most damage is done.
- Example: skill and luck are independent, but both determine whether a fund survives. Study only survivors and skill and luck become correlated.
- **Do not control for colliders** — and note that “control for” includes *sample selection*.

The one table to memorize

Structure	Shape	Association flows already?	Conditioning on middle node	Should you?
Chain (mediator)	$D \rightarrow M \rightarrow Y$	Yes, <i>causal</i>	blocks it	No
Fork (confounder)	$D \leftarrow X \rightarrow Y$	Yes, <i>spurious</i>	blocks it	Yes
Collider	$D \rightarrow C \leftarrow Y$	No	opens it	No

- Column 3 asks whether D and Y are *already* correlated through this structure; column 4, what conditioning does to that.
- Only the **fork** is worth conditioning on: it is the only structure where association flows and is *not* the effect you want.
- Mediators and colliders are both “leave alone” — for opposite reasons. Blocking a mediator deletes the answer; opening a collider invents one.
- Conditioning on a **descendant** of a collider partially opens it too. A proxy for a collider is still a collider.

4. Backdoor Paths and the Backdoor Criterion

Backdoor paths

- A **backdoor path** is a path connecting D and Y that begins with an arrow pointing *into* D .
- Compare: $D \rightarrow Y$ leaves D through an arrow pointing *out*, so it is a front-door — the causal path.
- Backdoor paths generate correlation without causation — “back doors” through which association leaks.
- Identification strategy in one sentence:

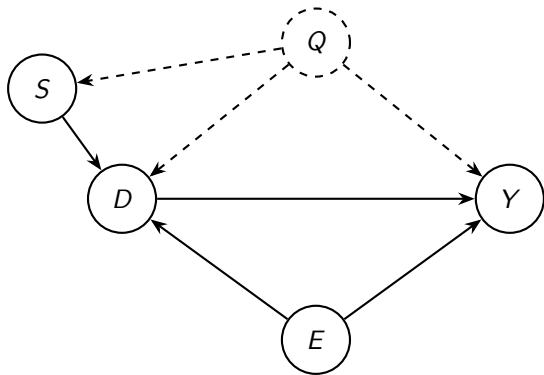
Leave every directed path from D to Y open.

Close every backdoor path.

- The **backdoor criterion**: a set of variables $\mathbf{Z}$ identifies the causal effect of D on Y if
 1. $\mathbf{Z}$ blocks every backdoor path from D to Y , and
 2. $\mathbf{Z}$ contains no descendant of D .
- Condition 2 is what rules out mediators and post-treatment colliders.

A DAG: Example

Effect of a bank **relationship** (D) on firm **investment** (Y).



- Q = firm quality (**unobserved**), S = firm size, E = local economic conditions.

Enumerate the paths

From the previous DAG:

1. $D \rightarrow Y$ — **directed**. Leave open; this is the estimand.
 2. $D \leftarrow Q \rightarrow Y$ — backdoor, open (fork at Q).
 3. $D \leftarrow S \leftarrow Q \rightarrow Y$ — backdoor, open (chain then fork).
 4. $D \leftarrow E \rightarrow Y$ — backdoor, open (fork at E).
- Conditioning on $\{Q, E\}$ blocks paths 2, 3, and 4. Identified.
 - But Q is **unobserved**. So the achievable set is $\{S, E\}$:
 - ▶ blocks path 3 (at S) and path 4 (at E),
 - ▶ leaves path 2 **open**.
 - Conclusion: the effect is **not identified** by controlling for observables.
 - That conclusion — reached before running anything — is the DAG earning its keep.

Simulation: the fork, and closing it

$Q \rightarrow D$, $Q \rightarrow Y$, $D \rightarrow Y$ with a true effect of 2.

```
set.seed(975)
n      <- 100000
qual   <- rnorm(n)                # firm quality (the confounder)
D      <- 0.6 * qual + rnorm(n)    # relationship lending
Y      <- 2 * D + 1.5 * qual + rnorm(n) # TRUE effect of D on Y is 2

b <- function(f) unname(coef(lm(f))[2])
round(c(naive = b(Y ~ D), control_Q = b(Y ~ D + qual)), 3)
```

```
naive control_Q
2.662      2.001
```

- Leaving the backdoor open: the estimate is too large by about 0.66.
- Blocking it at Q : we recover the truth.
- Here “blocking” just means putting Q in the regression.

Multiple valid strategies

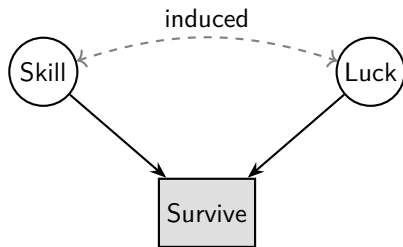
- Often several different sets satisfy the backdoor criterion.
- Any of them gives an **unbiased** estimate — so choose among them on **precision**.
- Rule of thumb: prefer the set that includes strong predictors of Y .
 - ▶ They do not change identification; they shrink residual variance and tighten standard errors.
- This is the legitimate version of “add controls to make the estimate sharper”.

5. Collider Bias

Colliders create correlation out of nothing

- The claim that surprises people: conditioning on a collider makes **independent** variables dependent.
- Intuition. Suppose a fund survives only if skill + luck is high enough.
 - ▶ Among survivors, a manager with low skill *must* have had high luck — otherwise they would not be in the sample.
 - ▶ So within survivors, skill and luck are **negatively** correlated, though they are independent in the population.
- The correlation is manufactured entirely by **who is in the sample**.

Survivorship as a collider



- Grey box = we conditioned on it (here, by studying only surviving funds).
- Skill $\rightarrow$ Survive $\leftarrow$ Luck is a collider. It was closed. We opened it.

Simulation: the surviving-fund fallacy

```
set.seed(975)
n      <- 100000
skill  <- rnorm(n)           # independent...
luck   <- rnorm(n)           # ...by construction
perf   <- skill + luck
surv   <- perf > quantile(perf, 0.80) # only the top 20% survive

round(c(cor_all      = cor(skill, luck),
        cor_survivors = cor(skill[surv], luck[surv])), 3)
```

```
cor_all cor_survivors
-0.003   -0.640
```

```
round(c(slope_all     = unname(coef(lm(perf ~ skill))[2]),
        slope_survivors = unname(coef(lm(perf[surv] ~ skill[surv]))[2])), 3)
```

```
slope_all slope_survivors
0.997     0.358
```

- Skill and luck: independent in the population, correlated -0.64 among survivors.
- The return to skill looks like 0.36 instead of 1.00 — understated.

This is not a hypothetical

- The same structure appears throughout empirical finance, usually unnoticed:
 - ▶ **Fund performance studies** on surviving funds.
 - ▶ **Compustat / Execucomp**: a firm appears only if it listed and survived. Listing is caused by size, quality, and opportunity — a collider.
 - ▶ **M&A studies** conditioned on **completed** deals: completion is caused by bidder synergies *and* target resistance.
 - ▶ **Rated firms only**: a firm is rated when it chooses to issue public debt.
 - ▶ **IPO samples**: going public is a common effect of quality and market conditions.
- Empirical finance treats these as a list of separate warnings — survivorship bias, sample coverage, deal completion. They are **one structural error**: conditioning on a collider.

Selection *is* conditioning

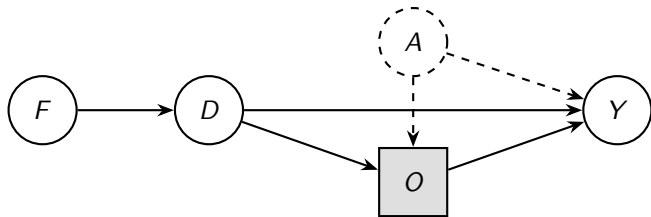
- You do not need to type a variable into a regression to condition on it.
- You condition on a variable whenever you
 - ▶ include it as a control,
 - ▶ stratify or match on it,
 - ▶ **drop observations** based on it,
 - ▶ or use a dataset that was **assembled** using it.
- The last one is the dangerous case, because it is invisible in your code.

Example. Your sample is the CRSP–Compustat merged file. A firm appears only if it (i) listed on a US exchange and (ii) filed accounts that got matched. Both are consequences of size, age, and quality — the same things driving your outcome. You conditioned on a collider without writing a single line of code that mentions it.

- Practical advice: write *how the sample was constructed* into the DAG, as nodes. Sample-construction filters are conditioning statements.

Discrimination and collider bias

- Gender discrimination is often denied like this: *condition on occupation and the wage gap shrinks or disappears* — within the same job, women are paid the same. **But what if occupation is itself a consequence of discrimination?**
- Suppose every woman is “treated” with discrimination and no man is. Firms respond in **two ways**: they pay her less, *and* they do not promote her — so she is paid less again.



F = female, D = discrimination, O = occupation, A = ability (unobserved), Y = wage

- Men and women are **equally productive**, so there is no arrow $F \rightarrow Y$. We are not after the effect of *female* on earnings; we are after the effect of **discrimination**.

What controlling for occupation actually does

Three paths run from D to Y :

1. $D \rightarrow Y$ — the **direct** effect. The causal path of interest.
 2. $D \rightarrow O \rightarrow Y$ — **mediation**, not a backdoor path: discrimination worsens the job, and the worse job pays less.
 3. $D \rightarrow O \leftarrow A \rightarrow Y$ — contains the **collider** O , so it is *closed* to begin with.
- Regress Y on D alone and you get the **total effect**: the direct effect plus the part mediated by occupational sorting.
 - Now add occupation, to “compare men and women in similar jobs.” Two things happen at once:
 - ▶ it **closes** the mediation channel $D \rightarrow O \rightarrow Y$ — reasonable, if the direct effect is what you want;
 - ▶ it **opens** path 3, because conditioning on a collider opens it. The control ironically introduces a new pattern of bias.
 - To shut path 3 again you would have to control for occupation **and ability** — and ability is unobserved. So this design **does not satisfy the backdoor criterion**.

The simulation

```
set.seed(975)
n          <- 10000
female     <- as.integer(runif(n) >= 0.5)    # half the population is female
ability    <- rnorm(n)                       # independent of gender; UNOBSERVED
discrimination <- female                      # all women experience discrimination
occupation <- 1 + 2*ability - 2*discrimination + rnorm(n)
wage       <- 1 - 1*discrimination + 1*occupation + 2*ability + rnorm(n)

b <- function(f) unname(round(coef(lm(f))["discrimination"], 3))
c(unconditional = b(wage ~ discrimination),
  add_occupation = b(wage ~ discrimination + occupation),
  add_occ_ability = b(wage ~ discrimination + occupation + ability))
```

unconditional	add_occupation	add_occ_ability
-2.983	0.591	-0.996

- Discrimination is **hard-coded**: -1 directly on wages, and -2 on occupation. True direct effect $= -1$; true total effect $= -1 + 1 \times (-2) = -3$.

Reading the three regressions

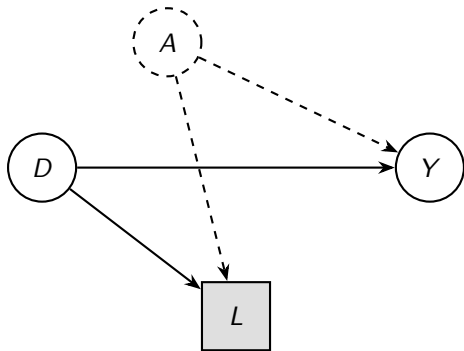
- **Unconditional** ≈ -3 . A large negative effect — the combination of the direct effect and the indirect effect through occupation. This is the *total* effect, and it is correct if that is what you want.
- **Add occupation** $\approx +0.6$. **The sign flips: it literally becomes positive.** We know this is wrong, because the direct effect was hard-coded at -1 . Conditioning on the collider O opened $D \rightarrow O \leftarrow A \rightarrow Y$, and the spurious path is strong enough to pervert the entire relationship.
- **Add occupation and ability** ≈ -1 . Now we recover the direct effect, because conditioning on A closes the path that conditioning on O opened.
- The rescue only worked because ability was **observable in the simulation**. In real data it is not — which is the whole point. “Control for occupation” is not the fix; it is the bug.

6. Bad Controls

The bad control problem

- A **bad control** is a variable that is itself affected by the treatment.
- Last lecture we previewed this and deferred the explanation. Here it is:
 - ▶ if $D \rightarrow X$, then X is a descendant of D ;
 - ▶ conditioning on it violates part 2 of the backdoor criterion;
 - ▶ it either blocks part of the causal effect (if X is a mediator) or opens a collider path (if X has another parent).
- The second case is the nasty one, because it can happen even when D was **randomly assigned**.

Randomization does not protect you



- D = credit supply shock (**randomly assigned**), Y = investment, L = leverage, A = firm quality (unobserved).
- Nothing points into D : there are **no backdoor paths**. The naive regression is already correct.
- But L is a collider on $D \rightarrow L \leftarrow A \rightarrow Y$. Control for leverage and you open it.

Simulation: how to ruin an experiment

```
set.seed(975)
n <- 100000
D <- rnorm(n)           # credit shock -- RANDOMLY ASSIGNED
A <- rnorm(n)           # firm quality -- unobserved
L <- D + A + rnorm(n)   # leverage -- affected by treatment
Y <- 1 * D + 3 * A + rnorm(n) # TRUE effect of D on Y is +1

b <- function(f) unname(coef(lm(f)))[2])
round(c(no_controls      = b(Y ~ D),
        control_L        = b(Y ~ D + L),
        control_L_and_A  = b(Y ~ D + L + A)), 3)
```

no_controls	control_L	control_L_and_A
0.993	-0.493	1.008

- Treatment was randomized. The true effect is +1. One control variable turns it into -0.5.

Reading the damage

- Without controls: D is randomized, no backdoor exists, estimate ≈ 1 . Correct.
- Adding L : opens $D \rightarrow L \leftarrow A \rightarrow Y$, and the estimate drops to -0.5 .
- Adding A as well: closes the newly opened path, restoring $+1$ — but A is unobserved in any real study.
- The lesson is not “leverage is a bad control.” It is:

*Whether a variable is a good control is not a property of the variable.
It is a property of its position in the graph.*

Bad controls in finance

Common post-treatment controls, and what they do:

- **Leverage / firm size / Tobin's Q** in a study of a shock that plausibly moves them.
- **Analyst coverage** when studying disclosure — disclosure changes coverage.
- **Deal completion** in M&A — completion is post-announcement.
- **Survival to $t + 1$** in any panel with attrition.
- **Post-treatment industry classification** for firms that change industries after treatment.

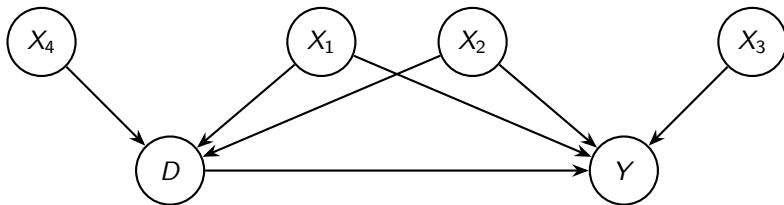
- The operational rule: **controls must be determined before treatment.**
- When in doubt, report the specification without the questionable control. If the estimate moves a lot, you have learned something — and you must explain which specification the DAG supports.

7. Neutral Controls and Precision

Not every control matters

- Some variables are neither confounders nor colliders. Conditioning on them changes nothing about bias.
- Two useful cases:
 - ▶ $X \rightarrow Y$ only (a predictor of the outcome, unrelated to D): **harmless and helpful** — it reduces residual variance and tightens standard errors.
 - ▶ $X \rightarrow D$ only (a predictor of treatment, unrelated to Y): **harmless but harmful in practice** — it absorbs variation in D and *inflates* standard errors.
- The second is the seed of the weak-instrument and “overcontrolling” problems you will meet in the IV lecture.

Precision: the setup



- X_1, X_2 are **confounders** — they point at both D and Y . Any valid set must include them.
- X_3 points at Y only. X_4 points at D only. Neither sits on a backdoor path, so **neither is needed** for identification.
- All three sets below therefore satisfy the backdoor criterion. They differ only in precision.

Precision: simulating three valid strategies

```
set.seed(975)
sim <- function() {
  n <- 1000
  x1 <- rnorm(n); x2 <- rnorm(n)      # confounders: -> D and -> Y
  x3 <- rnorm(n)                      # predicts Y only
  x4 <- rnorm(n)                      # predicts D only
  D <- 0.5*x1 + 0.5*x2 + 1.5*x4 + rnorm(n)
  Y <- 2*D + 2*x1 + 2*x2 + 4*x3 + rnorm(n)  # TRUE effect = 2
  b <- function(f) unname(coef(lm(f))[2])
  c(minimal = b(Y ~ D + x1 + x2),
    plus_x3 = b(Y ~ D + x1 + x2 + x3),
    plus_x4 = b(Y ~ D + x1 + x2 + x4))
}
r <- replicate(1000, sim())
round(rbind(mean = apply(r, 1, mean), sd = apply(r, 1, sd)), 4)
```

	minimal	plus_x3	plus_x4
mean	1.9999	2.0002	1.9962
sd	0.0735	0.0172	0.1299

Reading the precision results

- All three are **unbiased** — every mean is 2.000. Identification is settled; only variance differs.
- Adding X_3 , which predicts **only** Y , cuts the standard deviation from 0.074 to 0.017 — four times sharper. It soaks up residual variance without touching identification.
- Adding X_4 , which predicts **only** D , raises it to 0.130. It absorbs the variation in D you were relying on.
- **Practical guidance:**
 1. First use the DAG to find sets that identify the effect. That is about *bias*.
 2. Then, among those sets, prefer strong predictors of Y and avoid strong predictors of D alone. That is about *variance*.
- Never reverse the order. Choosing controls by what tightens the standard error is how sign flips get published.

8. What DAGs Cannot Do

Honest limitations

1. **DAGs do not test assumptions.** A DAG is an argument, not evidence. The data cannot tell you which way an arrow points.
2. **DAGs are non-parametric.** They tell you *whether* an effect is identified, never how large it is, nor the functional form, nor the estimator.
3. **DAGs are acyclic.** Genuine simultaneity — the equilibrium feedback that pervades finance — cannot be drawn. Time-indexing helps; it does not always suffice.
4. **DAGs say nothing about magnitude.** A technically open backdoor path with a tiny effect may matter less than a closed one. The graph is binary; the world is not.
5. **DAGs do not choose estimands.** *ATE* versus *ATT* is a potential outcomes question.

DAGs and potential outcomes are complements

Potential outcomes is better for	DAGs are better for
Defining estimands (ATE , ATT , ATU)	Choosing what to control for
Heterogeneous effects, δ_i	Spotting colliders and bad controls
Deriving estimators and their bias	Communicating a design in one picture
Inference and standard errors	Proving an effect is <i>not</i> identified

- Use the DAG to decide your specification; use potential outcomes to say what the coefficient means.

How to use this in your own work

1. Draw the DAG **before** the regression — including how the sample was built.
2. Mark unobserved variables with dashes. Be honest about what you cannot measure.
3. List every path from D to Y . Classify each: directed, or backdoor.
4. Find a set satisfying the backdoor criterion. If none exists using observables, **say so** — and go find a research design (IV, RDD, DiD) instead of more controls.
5. Among valid sets, pick for precision.

Wrapping Up

Summary [Part 1]

- A DAG is a picture of your causal assumptions; the **missing** arrows carry the content.
- **Association flows along all paths; causation flows only along directed paths.**
- Three structures, and the whole calculus follows from them:
 - ▶ **Chain** $D \rightarrow M \rightarrow Y$: open; conditioning closes it. Do not control for mediators.
 - ▶ **Fork** $D \leftarrow X \rightarrow Y$: open; conditioning closes it. Do control for confounders.
 - ▶ **Collider** $D \rightarrow C \leftarrow Y$: closed; conditioning **opens** it. Never control for colliders.
- **Backdoor criterion:** block every backdoor path, and condition on no descendant of D .

Summary [Part 2]

- **Collider bias is real and large.** Independent variables became correlated at -0.64 purely through sample selection; the return to skill fell from 1.00 to 0.36.
- **Selection is conditioning.** Dropping observations, or using a dataset assembled on a criterion, conditions just as surely as a regression control does.
- **Randomization does not immunize you:** a randomized treatment with one post-treatment control produced -0.5 when the truth was $+1$.
- Whether a control is good is a property of its **position in the graph**, not of the variable.
- Identify first (bias), then choose among valid sets (variance). Never the reverse.
- Next lecture: potential outcomes and DAGs both assumed we could condition our way to identification. What if we cannot? — **Unconfoundedness and matching.**

References

- Cunningham, S. *Causal Inference: The Remix*, Ch. 3.
- Pearl, J. (2009). *Causality: Models, Reasoning, and Inference*, 2nd ed. Cambridge University Press.
- Pearl, J. and D. Mackenzie (2018). *The Book of Why*. Basic Books.
- Cinelli, C., A. Forney, and J. Pearl (2024). A Crash Course in Good and Bad Controls. *Sociological Methods & Research* 53, 1071–1104.
- Angrist, J. and J.-S. Pischke (2009). *Mostly Harmless Econometrics*, §3.2.3. Princeton University Press.
- Elwert, F. and C. Winship (2014). Endogenous Selection Bias: The Problem of Conditioning on a Collider Variable. *Annual Review of Sociology* 40, 31–53.
- Kisgen, D. and P. Strahan (2010). Do Regulations Based on Credit Ratings Affect a Firm's Cost of Capital? *Review of Financial Studies* 23, 4324–4347.